

CivicTrack System

Phase 1: Project Proposal

Planning & Feasibility Analysis

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1. Introduction

Municipality service delivery is a vital responsibility of local authorities, especially in developing societies where problems are common. In such societies, residents are often confronted with problems such as damaged roads, water leaks, electricity outages, and dumping of refuse in undesignated places.

Although municipality service delivery is vital and essential in developing societies, most local authorities are still using conventional means of reporting issues that require residents to physically present themselves at local authorities' offices.

However, such reporting of issues may cause problems such as unnecessary transportation costs, lengthy waiting time, and lack of feedback regarding the status of issues reported.

In order to mitigate such problems and improve service delivery in local authorities, this project will propose a Community Issue Reporting and Tracking System known as CivicTrack. This system will enable residents of local authorities to electronically submit their issues and track their issues in real time.

2. Background of the statement problem:

Managing public service challenges including potholes, water leaks, interruptions in power, illegal dumping, and streetlight failures are an everyday problem with municipalities in many South Africa areas, including Johannesburg and Pretoria. Currently, many of municipalities relies on manual methods of reporting (phone calls, walk-ins), complaints due to the emails, logging systems that depend on paper and branch breakdowns in internal systems. These outdated methods promote inefficiencies, ineffective tracking, delayed responses, and a lack of transparency of the general populations.

Introduction to the core problem:

Due to outdated, disorganized and largely municipal service delivery in many South Africa cities encounter from significant operational inefficiencies. Infrastructure issues such as illegal dumping and damaged streetlights become regular throughout the people to deal with. These doesn't appear to be integrated digital electronic system that permits reporting, tracking, evaluation and monitoring of municipal service complaints, despite the urgency and prevalence of these obstacles. With the absence of a central organization, municipal departments encounter accountability challenges, communication difficulties, and operational bottlenecks.

DETAILED BREAKDOWN OF THE CORE PROBLEM

The core problem in the current system is that there are many separate and uncoordinated ways for citizens to report municipal service issues. Residents often report problems by calling municipal call centres, visiting local government offices in person, sending emails to different departments, or posting complaints on social

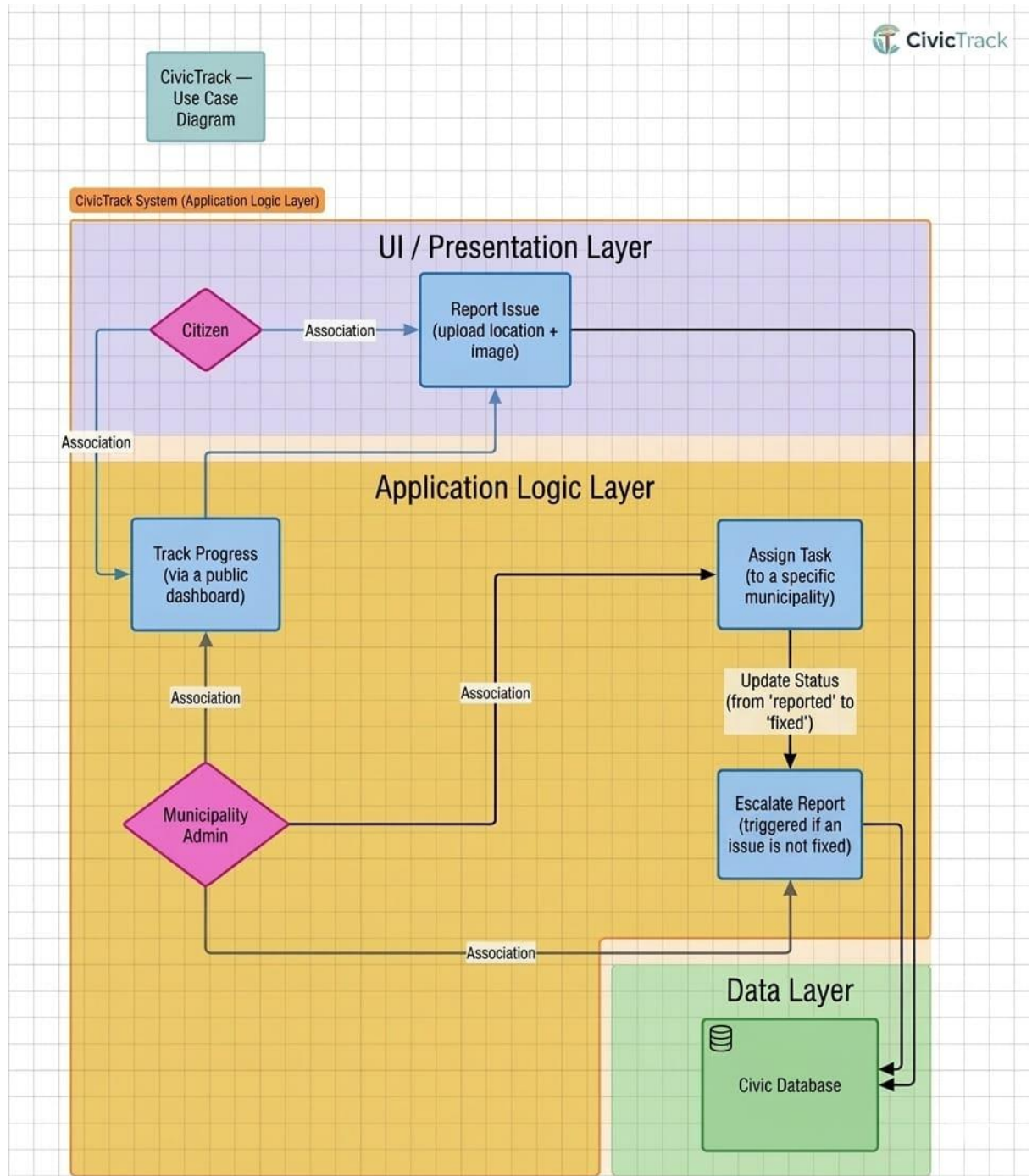
media. Because these channels are not connected to a single system, managing complaints becomes difficult. Some reports are written down manually by staff, emails may be sent to the wrong department or ignored, and phone calls may not always be recorded correctly. This situation can also lead to duplicate complaints being logged without anyone noticing.

In addition, the process of handling these issues lacks a structured workflow. Complaints are usually recorded manually by a call centre operator and then forwarded through internal emails or memos. Supervisors must manually assign technicians to handle the reported problems, and technicians later update supervisors about the progress. This manual process makes it difficult to assign tasks efficiently, determine the urgency of different issues, and monitor progress in real time. Another major problem is that citizens have no way to track the status of their complaints. In many cases, they are not given a reference number, do not receive updates about the progress of their report, and are not notified when the issue has been assigned or resolved. As a result, residents are often left uncertain about whether their concerns are being addressed, which can reduce trust in municipal service delivery.

3.Solutions

- Provide an online Report Issue form for easy problem submission.
- Allow users to select issue categories to route reports correctly.
- Enable location-based reporting using maps or address input.
- Enable photo uploads to confirm reported problems.
- Automatically allocate reports to appropriate municipal departments.
- Show problems on a color-coded map for rapid visual comprehension.
- Create a dashboard for users to monitor their submitted reports.
- Send notifications or alerts when issue status changes.
- Track and display average resolution times to improve accountability.

4. Architecture Clarity



References

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